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Male ZDF rats were tested weekly for the presence of hyperalgesia and tactile allodynia. Once allodynia was established, ZDF and lean rats were treated intravenously (i.v.) two times a week with an anti-LPA antibody (B3) at 10mg/kg or IgG at 10mg/kg, or saline. Paw tactile sensitivity was assessed as the primary indicator of efficacy. Results showed that The anti-LPA antibody, B3, was effective in increasing the paw withdraw threshold of allodynic rats when compared to both, IgG control or saline treated animals.
In summary, this study demonstrated efficacy of the murine anti-LPA antibody, B3, to reverse mechanical allodynia developed by animals in the ZDF rat model of type 2 diabetes.

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Efficacy of murine anti-LPA Ab in the ZDF rat model of type 2 diabetes

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1 SUMMARY

This efficacy study in Zucker diabetic fatty (ZDF) rats was conducted at University of California San Diego (UCSD San Diego, CA). The purpose of this study was to test the efficacy of the mouse anti-lysophosphatidic acid (LPA) antibody, B3, to ameliorate pain behavior in the ZDF rat model of type 2 diabetes.

Male ZDF rats were tested weekly for the presence of hyperalgesia and tactile allodynia. Once allodynia was established, ZDF and lean rats were treated intravenously (i.v) two times a week with an anti-LPA antibody (B3) at 10 mg/kg or control IgG (1015) at 10 mg/kg, or saline. Paw tactile sensitivity was assessed as the primary indicator of efficacy. Results showed that B3, was effective in increasing the paw withdraw threshold of allodynic rats when compared with both, 1015 or saline treated animals.

In summary, this study demonstrated efficacy of the murine anti-LPA antibody to reverse mechanical allodynia developed by animals in the ZDF rat model of type 2 diabetes.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is a key extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first monoclonal antibody targeting LPA (anti-LPA Ab). In preliminary studies, Lpath's anti-LPA mAb has demonstrated efficacy in ameliorating pain behaviors in diabetic, chemotherapeutic, arthritis and sciatic nerve injury induced-neuropathic pain. In the streptozotocin-induced model of diabetic neuropathy we have demonstrated that there is an up-regulation of LPA and that decreasing the levels of LPA with an anti-LPA antibody is sufficient to ameliorate tactile allodynia, a behavioral index of diabetes-induced neuropathic pain. Furthermore, therapy with anti-LPA Ab provides a long lasting sustained effect in accordance with the pharmacokinetic profile of the antibody. Our data demonstrates a role for LPA in diabetic neuropathy and suggests that anti-LPA Ab therapy may serve as a novel therapeutic strategy in the treatment of diabetes- induced neuropathies. To further evaluate the efficacy of the anti-LPA antibody in diabetes-induced neuropathic pain we used the Zucker diabetic fatty rat (ZDF) as a model for the study of neuropathic pain in type 2 diabetes (T2D), the most common form of the disease that represents more than 90% of all cases. Thus, the goal of this study was to test the efficacy of the mouse anti- LPA antibody, B3, to ameliorate pain behavior in the ZDF rat model of type 2 diabetes.

3 MATERIALS AND METHODS

3.1 Test Article and Control

Murine anti-LPA antibody (B3) Lot# HR 111266

Murine IgG2b (1015 control) Lot# HR91184

3.2 Study Design

Male ZDF (30) or lean (10) rats were tested weekly for the presence of hyperalgesia and tactile allodynia. Once allodynia was established, ZDF and lean rats were treated intravenously (i.v) two times a week with saline, anti-LPA Ab (B3) or 1015 at 10 mg/kg (see Table 1). Paw tactile sensitivity was assessed 3 days after each dose, as the primary indicator of efficacy for up to 5 doses (about 3 weeks). Blood was collected weekly (100 ul) to assay immunogenicity against the mouse antibody. Blood samples were sent to Lpath and analyzed for quantification of B3 and for the presence of anti-B3 antibodies. The study was terminated 3 days after the 5th dose.

Table 1. Study design

Group	Treatment	# rat/group	Rout/frequency
1	Lean (control) + saline	10	i.v 2X week
2	ZDF + saline	10	i.v 2X week
3	ZDF + B3 (10 mg/kg)	10	i.v 2X week
4	ZDF + 1015 (10 mg/kg)	10	i.v 2X week

3.3 Blood Collection and Analysis

Blood samples were collected in tubes containing potassium EDTA (K_2EDTA) predose (prior to dosing), and 24 hours after last dose each week for the duration of the study**.** Blood samples were centrifuged at approximately 5000 rpm for approximately 10 minutes at 4°C, and plasma was harvested. Plasma samples were immediately frozen in dry ice and stored at -70 ± 10°C until assayed for B3 and 1015 levels, the presence of anti-mouse antibody in rats (Anti-ADA) and for bioactive LPA.

B3 and 1015 antibody plasma concentrations were quantified using a direct-binding immunoassay (ELISA) method. Goat-anti mouse IgG (gamma-chain specific) UNLB antibody (Southern Biotech 1030-01, 1 mg/mL) was diluted 1:1000 in carbonate buffer (100 mM $NaHCO_3$, 33.6 mM Na_2CO_3, pH 9.5). 384 well plates were coated with 15 µL/well of this coating solution and incubated at 4°C overnight. The plates were then washed 4 times with PBS and blocked with 50 µL/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) for 1 hour at room temperature. Samples and standard were prepared on non-binding plates with enough volume to run in duplicate. The standard was prepared by diluting mouse anti-LPA antibody in blocking buffer (1%BSA, 0.1% Tween-20 in PBS) to the following dilutions: 900 ng/ml, 300 ng/ml, 100 ng/ml, 33.33 ng/ml, 11.11 ng/ml, 3.70 ng/ml, 1.23 ng/ml, 0.41 ng/ml, 0.14 ng/mL, 0.05 ng/mL, 0.02 ng/mL, and 0 ng/mL. The samples were diluted in blocking buffer 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490 and 15 µL of the standard/samples preparation were loaded to each well and incubated at room temperature for 1 hour. Plates were washed 4 times with PBS and 15 µL of 1: 10,000 blocking buffer diluted goat anti-mouse IgG HRP (Southern Biotech 1030-05) was added to each well and incubated for 1 hour at room temperature. Then, the plates were washed 4 times with PBS and developed using 15 µl/well TMB substrate at 4°C. After 5-20 minutes, the reaction was stopped with 1M H_2SO_4, 15 µl/well. The OD was measured at 450 nm. Data was analyzed using Graphpad Prism software. The standard ODs were graphed using a four parameter equation and this was used to calculate the mouse IgG in the samples. Excel software was used to calculate the µg/ml of mouse IgG in each sample by correcting the values for the dilution factor.

The presence of anti-ADA antibodies were detected by a direct-binding immunoassay (ELISA) method similar to the one described above. B3 antibody was diluted to 1 µg/mL in carbonate buffer (100 mM $NaHCO_3$, 33.6 mM Na_2CO_3, pH 9.5). 384 well plates were coated with 15µL/well of coating antibody solution and incubated at 4°C overnight. The next morning, plates were shook out and 25 µL/well blocking buffer (1%BSA, 0.1% Tween-20 in PBS) was added and incubated for 1 hour at room temperature. Plates were washed 4 times with PBS, and 15 µL of diluted test samples per well was added in duplicate and incubate for 1 hour at room temperature. Rat plasma samples were diluted with blocking buffer to 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490. Then, plates were washed 4 times with PBS and 15 µL of 1:5,000 blocking buffer diluted goat anti-rat IgG HRP (Jackson 112-035-062) was added to each well and incubate for 1 hour at room temperature. Finally, the plates were washed 4 times with PBS and developed with 15 µl/well TMB substrate at 4°C. After 5 minutes, the reaction was stopped with 1M H_2SO_4, 15 µl/well and the OD was measured at 450 nm. Data was analyzed using Graphpad Prism software.

Bioactive LPA was detected using The CHO-K1 EDG2 (LPA_1 receptor) β-arrestin clone (PathHunter™) that uses Enzyme Fragment Complementation technology in which two weakly complementing fragments of the B-galactosidase (β-gal) enzyme are expressed in stably transfected cells. One fragment of the β-gal, termed the enzyme acceptor (EA), is fused to the C-terminus of β-arrestin2. The complementing fragment of β-gal, termed the ProLink™ tag, is expressed as a C-terminal fusion protein with the LPA_1 receptor. Upon activation, the LPA_1 receptor is phosphorylated, providing a binding site for β-arrestin. The interaction of β-arrestin and the LPA_1 receptor forces the interaction of Prolink and EA, thus allowing complementation of the two fragments of β-gal and the formation of functional enzyme capable of hydrolyzing substrate and generating a chemiluminescent signal. Briefly, CHO-K1 EDG2 cells were plated at and incubated at 37°C/5% CO_2. After 24 hours, plates were starved with reduced serum media at 37°C for another 24 hours. LPA standard curves were prepared by titrating LPA (0 – 60 µM) in F12 media containing 1mg/mL BSA onto the plate to determine low and maximal signal (dotted lines on Figure 2). Standards and samples were added to the plate and incubated at room temperature for 90 minutes. The plates were washed, incubated with the detection reagent in the dark for 90 minutes and read on a standard luminescence plate reader.

3.4 Tactile Response Threshold

Rats were transferred to a testing cage with a wire mesh bottom and allowed to acclimate. Von Frey filaments (Stoelting, Wood Dale IL) were used to determine the 50% mechanical threshold for foot withdrawal. A series of filaments, starting with one possessing a buckling weight of 2.0g, were applied in sequence to the plantar surface of the right hindpaw with a pressure that causes the filament to buckle. Lifting of the paw was recorded as a positive response and the next lightest filament chosen for the next measurement. Absence of a response after 5 seconds prompted use of the next filament of increasing weight. This paradigm was continued until four measurements were made after an initial change in the behavior or until five consecutive negative (given the score of 15g) or four positive (score of 0.25g) scores occurred. The resulting sequence of positive and negative scores was used to interpolate the 50% response threshold. Only rats with a 50% tactile response threshold below 6 g are considered allodynic and brought forward for drug testing.

3.5 Data Analyses

All data was generated and collected at the laboratory of Dr. Nigel Calcutt at UCSD. Descriptive statistics was used to calculate median, mean, standard deviation (SD) and standard error of the mean (SEM) using Microsoft Excel. An excel spread sheet containing all data and analysis was sent via e-mail to Lpath (Table 2). GraphPad Prism software (version 6.0, Graph Software inc, La Jolla, CA) was used to graph and analyze (ANOVA) the data.

4 RESULTS

4.1 Body Weight

ZDF diabetic rats had a greater weight than control lean rats by the end of the study at 20 weeks of age. The mean (±SEM) body weight for diabetic ZDF rats at conclusion of the study (450 ± 10 g) was significantly higher than the corresponding mean (±SEM) body weight of the lean control group (351± 5 g) (****p <0.0001 by one-way ANOVA with Bonferroni *post-hoc* analysis). There was no significant difference in weight between, saline, B3 antibody and 1015 control-treated ZDF rats (Table 2).

4.2 Blood Glucose

Blood glucose concentration increased approximately 4-fold in diabetic ZDF rats compared with the control lean group (Table 2). The mean (±SEM) blood glucose for diabetic ZDF rats at the end of the study (414±36 dl/ml) was significantly higher than the corresponding mean (±SEM) blood glucose of the lean control group (113±3 dl/ml) (****p <0.0001 by one-way ANOVA with Bonferroni *post-hoc* analysis). There was no significant difference in blood glucose between, saline, B3 antibody and 1015 control-treated ZDF rats. Similar results were observed when the HbA1c (glycosylated hemoglobin A1c) test was used to measured average level of blood sugar over the previous 3 months (Table 2).

4.3 Antibody Concentration at Termination

The antibody concentration in plasma of animals treated with B3 antibody or 1015 control was as expected for the administered dose of 10 mg/kg twice a week (Table 3). No anti-B3 or anti-1015 antibodies were detected (Appendix 1).

4.4 Mechanical Allodynia

Mechanical allodynia (defined as 50% response threshold < 6g) was fully developed in the hindpaws of diabetic ZDF rats by 17 weeks of age (Baseline on Table 2 and Day 0 on Figure 1).

Once allodynia was confirmed, animals were randomized in treatment groups and dosing was started as described in Table 1. Mechanical allodynia was measured on the 3rd day after each dose for 5 doses (Table 2; Figure 1). Results showed that ZDF diabetic rats treated with B3 antibody recovered from established allodynia when compared with saline (vehicle) or 10 mg/kg 1015 (negative control). The latter group remained allodynic (PWT <6g) up to the end of the study (****p <0.0001 v. 1015 by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis).

4.5 Bioactive LPA

The levels of bioactive LPA (defined as the amount of LPA present in plasma able to activate the LPA_1 receptor in the DiscoveRX reporter cell assay) were measured in the plasma of non-diabetic (control) and ZDF Vehicle, 1015 and B3 treated animals. We demonstrated that there was an increase in bioactive LPA in the diabetic animals (*p <0.05 for saline, and ****p <0.0001 for 1015 treated rats) which was significantly reduced in animals treated with B3 (**p <0.01, vs. 1015 by one-way ANOVA with Bonferroni *post-hoc* analysis. (Figure 2).

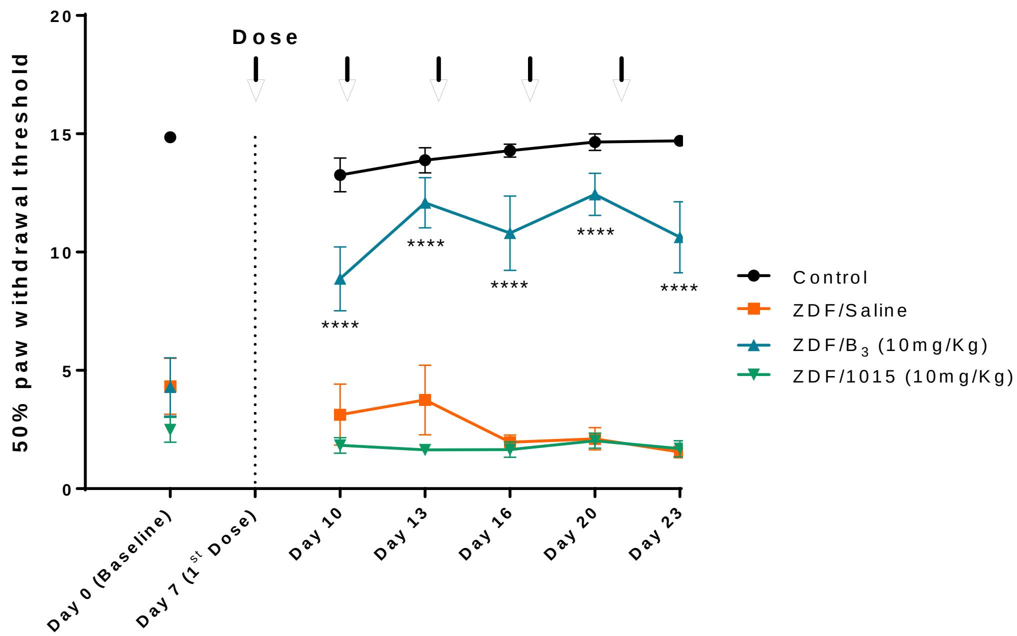
Table 2. 50% Paw withdrawal threshold (PWT) and terminal Body Weight, Blood Glucose and HbA1c levels in Lean (control) and ZDF rats dosed iv two times a week with Saline, 10 mg/kg B3 or 10 mg/kg IgG control (1015)

Model	ZDF rats born 3 Aug 2012								
date	27-Nov-12	7-Dec-12	10-Dec-12	13-Dec-12	17-Dec-12	20-Dec	20-Dec	20-Dec	20-Dec
designation	Baseline	dose #1+3 days	dose #2+3 days	dose#3+3 days	dose #4+3 days	dose #5 +3 days	terminal	terminal	terminal
test	50% PWT	50% PWT	50% PWT	50% PWT	50% PWT	50% PWT	body weight	blood glucose	HbA1c
control									
1.0	15.00	13.50	10.63	12.68	15.00	15.00	375	122	5.0
1.1	15.00	11.14	15.00	15.00	15.00	13.50	330	121	4.8
1.2	15.00	13.56	15.00	15.00	15.00	15.00	329	118	4.7
2.0	15.00	10.05	15.00	13.50	11.46	15.00	365	115	6.3
2.1	15.00	15.00	15.00	15.00	15.00	15.00	353	120	4.7
2.2	13.50	15.00	15.00	13.50	15.00	15.00	330	123	4.9
3.0	15.00	15.00	11.17	14.11	15.00	15.00	372	104	4.6
3.1	15.00	9.37	13.50	14.11	15.00	13.50	361	100	4.7
4.0	15.00	15.00	15.00	15.00	15.00	15.00	348	108	5.1
4.1	15.00	15.00	13.50	15.00	15.00	15.00	350	97	4.7
Median	15.00	14.28	15.00	14.56	15.00	15.00	352	117	4.8
Mean	14.85	13.26	13.88	14.29	14.65	14.70	351	113	5.0
SD	0.47	2.24	1.69	0.84	1.12	0.63	17	10	0.5
SEM	0.15	0.71	0.53	0.27	0.35	0.20	5	3	0.2
ZDF + saline									
5.0	2.40	1.05	1.55	1.70	0.52	0.88	456	503	9.8
5.1	2.18	0.97	1.70	1.55	0.64	1.05	469	413	9.1
6.0	1.92	2.28	1.75	1.87	1.12	1.85	445	445	11.1
6.1	2.18	1.62	1.04	1.92	3.58	1.29	423	415	10.0
7.0	9.67	2.59	2.70	2.76	4.77	2.64	501	143	4.8*
7.1	3.43	2.28	0.50	1.52	1.22	2.74	382	566	10.8
8.0	3.24	14.49	2.97	3.51	3.85	1.07	496	452	10.1
8.1	1.79	1.31	1.07	0.25	2.18	1.56	449	344	10.0
9.0	12.68	3.85	15.00	3.30	2.33	1.87	458	397	8.2
9.1	2.81	0.90	9.17	1.24	0.90	0.42	419	464	10.5
Median	3.03	1.95	1.73	1.79	1.70	1.43	453	430	10.0
Mean	4.33	3.13	3.75	1.96	2.11	1.55	450	414	10.0
SD	3.73	4.09	4.66	0.98	1.50	0.76	32	113	0.9
SEM	1.18	1.29	1.47	0.31	0.47	0.24	10	36	0.3
ZDF + B3 (10 mg/kg iv)									
10.0	2.85	4.78	8.07	12.99	11.17	6.15	461	391	13
10.1	8.70	7.66	11.85	11.99	13.50	9.06	430	392	13
11.0	1.96	7.40	15.00	10.18	8.13	12.68	350	427	13
11.1	1.69	2.72	8.38	10.34	8.78	0.25	501	190	-
12.0	10.71	13.50	15.00	15.00	13.50	15.00	412	400	13
12.1	9.01	4.11	15.00	13.50	15.00	15.00	414	450	12.3
13.0	3.24	15.00	15.00	15.00	15.00	15.00	473	365	11.2
13.1	0.42	10.32	10.71	15.00	15.00	13.50	456	384	9.0
14.0	2.09	13.50	15.00	2.81	15.00	9.37	433	448	10.8
14.1	1.32	9.67	6.97	1.22	9.36	10.32	489	336	7.1
Median	2.47	9.67	13.43	12.49	13.50	11.50	445	392	12.3
Mean	4.30	8.87	12.09	10.80	12.44	10.63	442	378	11.4
SD	3.90	4.27	3.35	4.97	2.82	4.73	44	75	2.1
SEM	1.23	1.35	1.06	1.57	0.89	1.50	14	24	0.7
ZDF + 1015 (10 mg/kg iv)									
15.0	2.83	2.75	1.80	1.65	1.43	0.83	474	199	8.3
15.1	2.95	0.90	1.15	1.87	0.57	0.62	452	416	13
16.0	1.60	0.90	0.42	1.80	3.58	1.75	471	406	12.0
16.1	6.68	3.06	2.59	4.04	3.24	3.42	433	358	13
17.0	2.00	3.36	1.85	1.41	3.13	3.33	407	466	13
17.1	0.72	0.25	0.59	0.25	1.69	0.78	382	403	13
18.0	2.79	1.65	1.53	0.90	2.44	2.28	456	383	10.9
18.1	1.59	1.58	2.00	1.53	1.92	0.62	434	405	10.4
19.0	2.83	1.37	2.33	2.13	1.30	2.00	428	386	-
19.1	0.88	2.50	2.05	0.90	0.99	1.22	432	457	10.1
Median	2.40	1.62	1.88	1.59	1.81	1.49	434	404	12.0
Mean	2.49	1.83	1.64	1.65	2.03	1.69	436	390	11.5
SD	1.69	1.04	0.72	1.01	1.03	1.07	29	75	1.7
SEM	0.53	0.33	0.23	0.32	0.32	0.34	9	24	0.6

Table 3. Antibody Concentration at End of Study

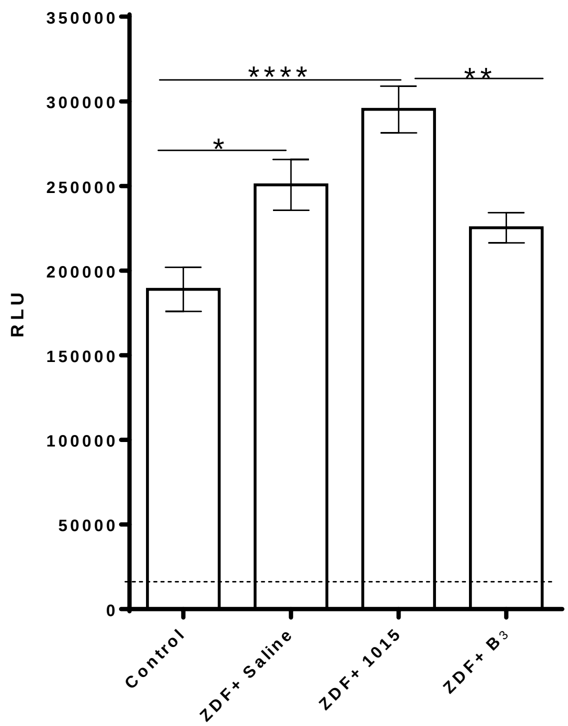
Terminal mAb Concentration, µg/ml					
Anti-LPA (B3 mAb) group			IgG control (1015 mAb) group		
Sample ID	Mean	SDev	Sample ID	Mean	SDev
10.0	115	32	15.0	181	1
10.1	117	5	15.1	167	15
11.0	130	14	16.0	174	11
11.1	174	2	16.1	190	44
12.0	65	1	17.0	367	48
12.1	93	7	17.1	154	9
13.0	110	4	18.0	176	7
13.1	129	0	18.1	174	3
14.0	103	7	19.0	143	6
14.1	85	5	19.1	148	8
Mean	**112**		**Mean**	**188**	
Sdev	**29**		**Sdev**	**65**	

Figure 1. B3 Ab reduces neuropathic pain in Zucker diabetic fatty (ZDF) rats.



Tactile responses, represented as 50% Paw Withdrawal Threshold (PWT), in control (black circles) and ZDF rats receiving intravenous injection of saline (orange square) or 10 mg/kg B3 (blue triangle), or 1015 control (green triangle) .
****p <0.0001 v. 1015 by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean ±SEM, n = 10/group.

Figure 2. Bioactive LPA



CHO-K1 EDG2 cells were plated as recommended and starved for 24 h. The plates were washed, and developed as recommended by the manufacture. Plasma samples from, control and ZDF rats receiving intravenous injection of saline or 10 mg/kg B3 or 1015 control were added to the plate and processed as described in methods. *p <0.05, **p <0.01 vs control, ****p <0.0001 vs 1015 by one-way ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean ±SEM, n = 10/group.

4.6 CONCLUSION

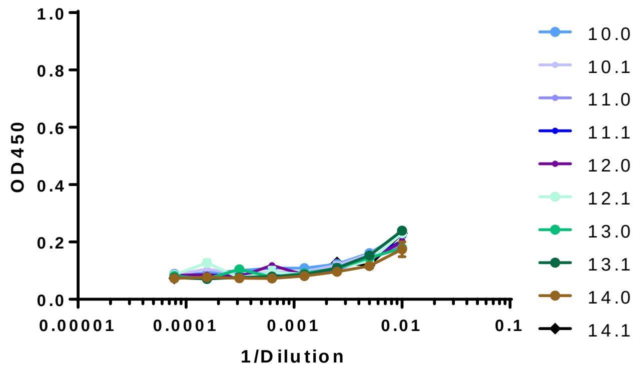
Therapeutic intervention with the anti-LPA antibody, B3 at 10 mg/kg decreased the circulating levels of bioactive LPA and alleviated tactile allodynia in the ZDF rat model of type 2 diabetes

4.7 **DATA** ARCHIVE

Lpath notebook #137 pp.57-63; #229 pp.134-136; #248 pp.83-86

5 APPENDICES

Appendix 1 Detection of anti-B3 antibodies by a direct-binding immunoassay



Anti-B3 antibody titers. B3 antibody was diluted to 1 µg/mL to coat plates and the ELISA was carried out as described in the methods. Rat plasma samples from B3 treated group were diluted with blocking buffer to 1:10, 1:30, 1:90, 1:270, 1:810, 1:2430, 1:7290, 1:21870, 1:65610, 1:196380 and 1:590490. Data was analyzed using Graphpad Prism software.